



DEPARTMENT OF COMPUTER SCIENCE

COMP338 - Artificial Intelligence

Course Project 1

Due date: May 26, 2026 11:59 PM

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Project Description:

Write your own implementation of *Genetic Algorithm* to generate an exam schedule with the following constraints:

- No student should have two exams at the same time
- A student should preferable not have two exams in the same day
- A student should not have more than two exams in the same day
- A student should not have four exams in two days in a row
- The schedule should use the minimum number of exam days
- A day can only fit three exam time slots

The dataset:

A sample dataset for 95 students registered in 22 different courses is attached, the dataset includes a list of registered courses for each student. Use that dataset to test your implementation.

Note: the dataset includes a sample schedule that violates the constraints, use it to test your fitness function.

Notes:

- You need to implement the algorithm on your own, using any programming language you prefer, YOU CANNOT USE existing libraries. This is to help you understand the very details of the algorithm.
- Formulating the fitness function can be the most critical part of the implementation, give it enough attention in your report.
- Convergence rate of Genetic Algorithm heavily depends on the tuning of its setting parameters. When implementing the algorithm, you will need to try the algorithm for many times while tuning the parameters. Draw a plot to show the convergence rate for

each of the proposed setting parameters. I suggest you have your program calculate and store the convergence rate in a file

What to submit:

1. Your source code
2. Report containing the following:
 - Background on Genetic Algorithms: explain its steps and its setting parameters
 - Make sure you clearly explain the problem formulation, you need to explain how you represent the solution, what does the chromosome represent, what does the gene represent.
 - explain the fitness function
 - Include and explain the code, each stage of the Genetic Algorithm.
 - Parameter tuning analysis: as mentioned earlier, make sure you include a plot to show the effect of the tuning on the convergence rate
 - References: make sure you quote every reference or website you read, the libraries you used.
 - Screenshots for the run
 - Results and discussion

Submission guidelines:

- The project is a teamwork of two students
- Submit through Ritaj as a reply to the assignment message ONLY, submissions through other means will not be accepted
- Deadline is announced in the Ritaj assignment message
- Plagiarism will not be tolerated
- Chatbot misuse will not be tolerated
- Prepare for a discussion, it will have a significant impact on your score, and if you don't attend the discussion your project will not be graded.

Best Luck